Lab 3.1 – Storage Account, RBAC and SAS

Module M7 · Estimated time: 45–60 minutes

Scenario

A business team needs to upload operational files into Azure Blob Storage. Engineers must access the data through Microsoft Entra ID, while an external reviewer should receive temporary read-only access to one specific file—not the entire container.

Objective

- Create and inspect a GPv2 Storage Account.
- Access Blob Storage using Microsoft Entra ID and RBAC.
- Upload the supplied Day 3 files to `raw`.
- Generate a read-only, one-hour user-delegation SAS for one blob.
- Prove that the SAS can read the blob but cannot list the container.
- Configure a lifecycle rule to move blobs in `raw` to Cool after 30 days.

Step 1 – Confirm your Storage Account

```
$RG = "<your-shared-resource-group>"
$ST = "stuser01ordersd3"
$CONTAINER = "raw"

az storage account show `
  --resource-group $RG `
  --name $ST `
```

```
  --query
"{name:name,kind:kind,sku:sku.name,location:location}"
`
  --output table
```

Expected: your Day 3 Storage Account is returned as a `StorageV2` account with Standard LRS.

```
PS C:\Users\LabsKraft> az storage account show `
>>    --resource-group "Kraft-Internal" `
>>    --name "stuser01ordersd3" `
>>    --query "{name:name,kind:kind,sku:sku.name,location:location}" `
>>    --output table
Name              Kind       Sku           Location
----------------  ---------  ------------  ------------
stuser01ordersd3  StorageV2  Standard_LRS  centralindia
PS C:\Users\LabsKraft>
```

Step 2 – Verify Blob data access

```
az storage container list `
  --account-name $ST `
  --auth-mode login `
  --output table
```

Expected: `raw` and `processed` are listed.

If you receive an authorization error, verify that your signed-in identity has Storage Blob Data Contributor on the Storage Account. Management-plane Contributor alone does not grant Blob data access.

```
PS C:\Users\LabsKraft> az storage container list `
>>    --account-name "stuser01ordersd3" `
>>    --auth-mode login `
>>    --output table
Name                                  Lease Status    Last Modified
------------------------------------  --------------  -------------------------
app-package-fn-user01-orders-d3-b41c6be               2026-11-05T11:09:53+00:00
azure-webjobs-hosts                                   2026-11-05T11:10:37+00:00
azure-webjobs-secrets                                 2026-11-05T11:11:02+00:00
processed                                             2026-11-05T10:35:08+00:00
raw                                                   2026-11-05T10:34:54+00:00
PS C:\Users\LabsKraft>
```

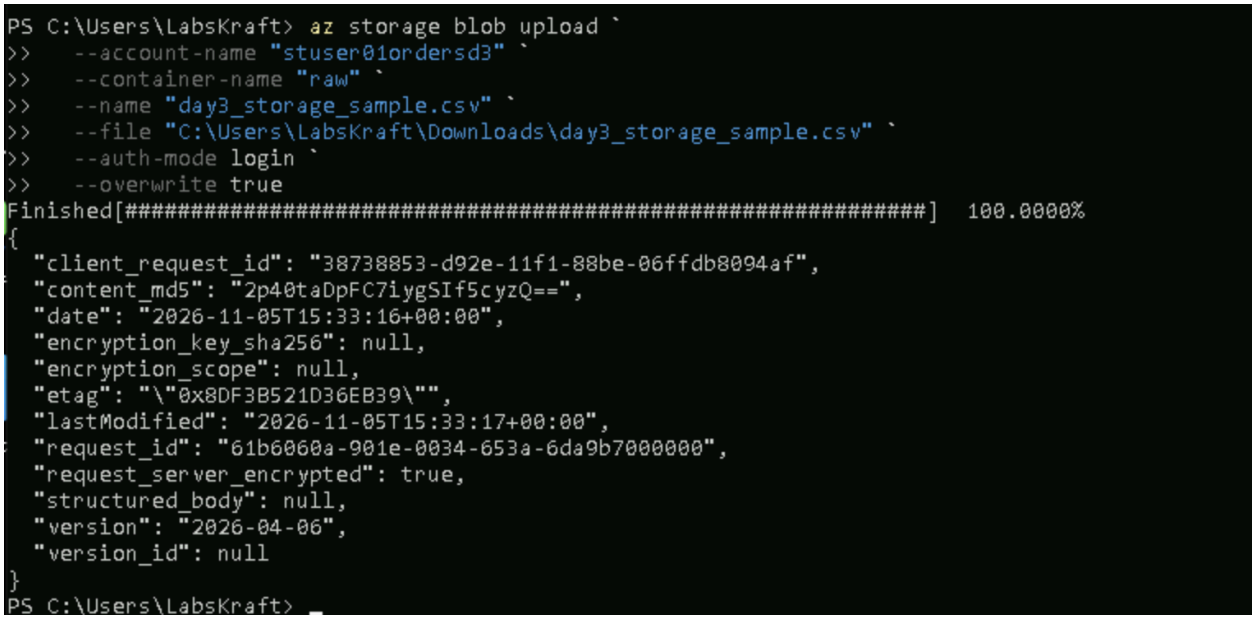
Step 3 – Upload the supplied training files

Use the existing files supplied in the Day 3 Artifacts folder. Do not create replacement datasets.

For each supplied file, run:

```
az storage blob upload `
  --account-name $ST `
  --container-name $CONTAINER `
  --name "<file-name>" `
  --file "<local-file-path>" `
  --auth-mode login `
  --overwrite true
```

Replace the placeholders with the actual supplied filename and its local path.

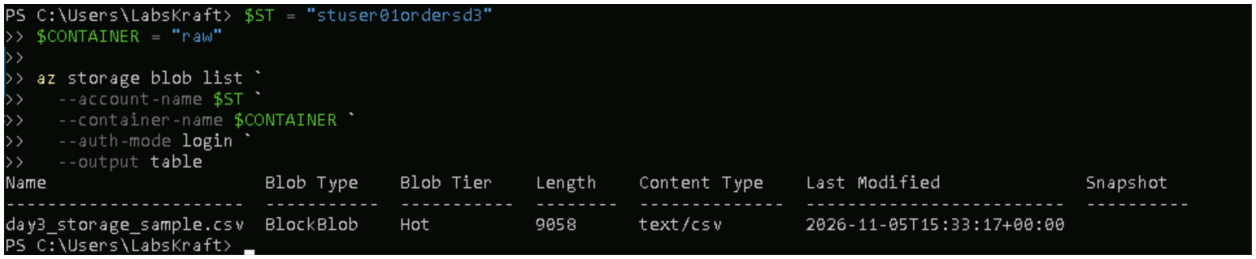


Verify:

```
az storage blob list `
```

```
  --account-name $ST `
  --container-name $CONTAINER `
  --auth-mode login `
  --output table
```

Expected: the uploaded files appear under `raw`.



Step 4 – Generate a user-delegation SAS

Choose one uploaded file:

```
$BLOB = "<uploaded-file-name>"
$EXPIRY =
(Get-Date).ToUniversalTime().AddHours(1).ToString("yyyy
-MM-ddTHH:mmZ")
```

Generate a read-only SAS:

```
$SASURL = az storage blob generate-sas `
  --account-name $ST `
  --container-name $CONTAINER `
  --name $BLOB `
  --permissions r `
  --expiry $EXPIRY `
  --auth-mode login `
  --as-user `
  --https-only `
```

```
  --full-uri
```

Display the generated URL:

```
$SASURL
```

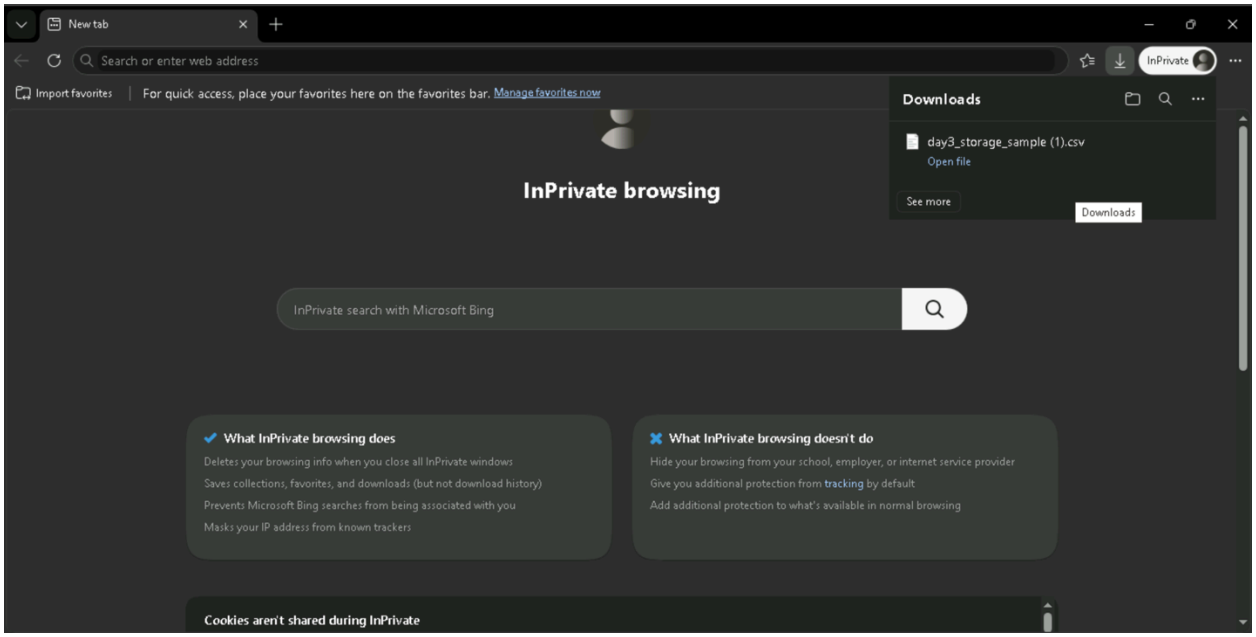
This creates a user-delegation SAS using your Entra identity. Treat the URL as a credential: do not paste it into tickets, screenshots or shared chat.



Step 5 – Validate the SAS permissions

Test A – Read the individual blob

1. Copy the generated SAS URL.
2. Open it in a private browser window.
3. Expected: the browser can access the selected blob, subject to its file type and browser handling.



Test B – Attempt container listing

Use the same SAS token against the container listing endpoint:

```
https://<storage-account>.blob.core.windows.net/raw?restype=container&comp=list&<sas-token>
```

Expected: the request fails because the SAS was created for one blob with read permission only. It does not grant container-list permission.

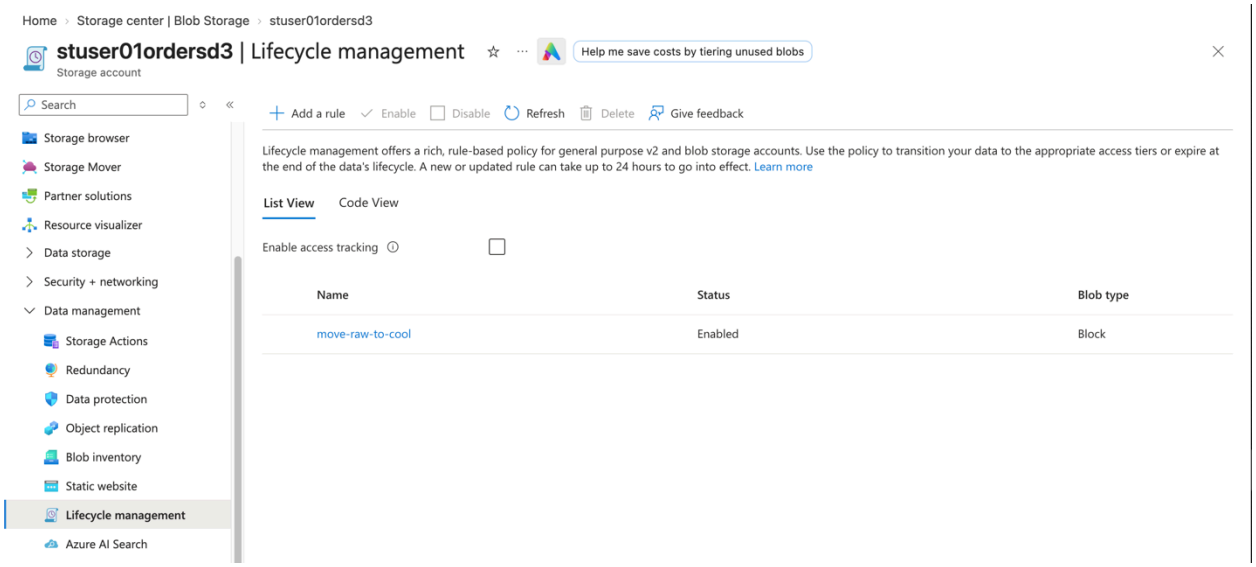
Do not add list permission to make this test pass. The failure is the expected security result.



Step 6 – Configure the Blob lifecycle rule

1. Open your Storage Account in Azure Portal.
2. Navigate to Data management → Lifecycle management.
3. Select Add a rule.
4. Configure:
    - Rule name: `move-raw-to-cool`
    - Rule scope: Limit blobs with filters
    - Blob type: Block blobs
    - Blob prefix: `raw/`
    - Base blobs: Move to Cool storage after 30 days since last modification
5. Save the rule.

Expected: the lifecycle rule is visible and enabled.



This configures a policy; it does not move the uploaded files immediately. The 30-day condition must be met before the action occurs.

Step 7 – Acceptance criteria

Validation	Expected result
Storage Account	GPv2, Standard LRS
Entra authentication	Blob commands succeed with `--auth-mode login`
File upload	Supplied files visible in `raw`
User-delegation SAS	Read access to selected blob succeeds
Container listing	Fails using the blob-level read-only SAS
Lifecycle rule	`raw/` blobs configured to move to Cool after 30 days

Production takeaway: Use Entra ID and narrowly scoped data roles for normal workforce access. Use short-lived SAS tokens when temporary external access is required. Lifecycle policies help control storage costs without requiring manual file movement.